

Comparative Evolutionary Analysis of the BRCA1 Gene Across Mammalian Species Using Bioinformatics Tools

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1. Abstract

The **BRCA1 (Breast Cancer 1)** gene plays a very important role in maintaining the stability of genome by playing role in DNA repair, cell cycle regulation, and the preservation of genetic integrity. Any kind of mutations in this gene are strongly associated with an increased risk of hereditary breast and ovarian cancers, making it an important subject of genetic and evolutionary research. The present study aimed to investigate the evolutionary conservation of the BRCA1 gene among selected mammalian species using freely available bioinformatics tools.

Reference mRNA sequences of the BRCA1 gene were retrieved from the NCBI RefSeq database for *Homo sapiens*, *Macaca mulatta*, *Canis lupus familiaris*, *Mus musculus*, and *Rattus norvegicus*. Pairwise sequence comparisons were performed using BLASTN to evaluate nucleotide similarity among species. Multiple sequence alignment was carried out using the MUSCLE algorithm in MEGA, followed by construction of a Neighbor-Joining phylogenetic tree with bootstrap analysis to infer evolutionary relationships.

The comparative analysis demonstrated that the human BRCA1 sequence shared the highest nucleotide similarity with *Macaca mulatta*, while lower similarities were observed with *Canis lupus familiaris*, *Mus musculus*, and *Rattus norvegicus*. The phylogenetic analysis grouped human and rhesus macaque together, whereas mouse and rat formed a separate rodent cluster, with dog occupying an intermediate position. These findings are consistent with the established evolutionary relationships among mammalian species and indicate that the BRCA1 gene has remained highly conserved during mammalian evolution.

This study demonstrates that freely available bioinformatics resources can be effectively used to investigate gene conservation and evolutionary relationships. The project also highlights the usefulness of computational approaches in molecular biology education and provides a foundation for future studies involving protein sequence analysis, conserved domain identification, and functional genomics.

2. Introduction

Every cell in our body contains DNA that stores genetic information. Every day the cells are exposed to DNA damage due to UV rays, radiation, harmful chemicals or cellular metabolism etc. Therefore, if these damages are not repaired efficiently can cause mutations in DNA that can lead to cancer and other genetic disorders.(Wu et al., 2010; Zhong et al., 2023)

To protect DNA from this, cells have different DNA repair mechanisms, For example:

- Base Excision Repair (BER)
- Nucleotide Excision Repair (NER)
- Mismatch Repair (MMR)
- Double-Strand Break Repair (Homologous Recombination aur Non-Homologous End Joining)

BRCA1 gene mainly works on **Homologous Recombination (HR)** pathway and accurately repairs double-strand DNA breaks.

BRCA1 (Breast Cancer 1) is a tumour suppressor gene that is present on chromosome 17 (17q21) which is responsible for synthesizing BRCA1 protein.

Major functions of BRCA1 protein:

- DNA repair
- Cell cycle regulation
- Maintain genome stability
- Detection of damaged DNA
- Controls cell division

Any type of mutation in BRCA1 gene increases the risk of hereditary breast cancer and ovarian cancer. Normally BRCA1 is responsible for repairing DNA damage but when there is a mutation in this gene causes no DNA damage repair which leads to accumulation and uncontrolled proliferation of cells, eventually causing the development of tumour. Hence, BRCA1 gene has its own importance in medical genetics. (Fu et al., 2022; Gorodetska et al., 2019; Pfeiffer et al., 2017)

Evolutionary conservation of a gene means that the gene has gone through very little amount of changes during millions of years of evolution and has been conserved since. Genes that play an important function necessary for survival can not be compromised and do not tolerate mutations. Therefore, they are conserved by Natural Selection. BRCA1 is one of those genes that's the reason in many mammalian species like dog, monkey, mouse, rat have very similar sequence of BRCA1 gene. (Burk-Herrick et al., 2006)

Comparative genomics is the process in which genes or genome of different species are compared for the following reasons:

- To identify which of the species are closely related.
- Which genes have been conserved.
- To evaluate the changes during evolution.

In this project we are going to compare BRCA1 gene in 5 different mammalian species.

The selected mammalian species share a common evolutionary ancestor but represent different evolutionary lineages, making them suitable for comparative genomic analysis. The purpose of species selection was to compare evolutionary distance.

Mammals that I selected were:

- Human (*Homo sapiens*)
- Rhesus macaque (*Macaca mulatta*)
- Dog (*Canis lupus familiaris*)
- Mouse (*Mus musculus*)
- Rat (*Rattus norvegicus*)

These mammalian species are commonly used in biomedical research and their BRCA1 sequences were available in NCBI database.

Bioinformatics is an interdisciplinary field that integrates biology, computer science, mathematics, and statistics to analyze biological data. (*Research Journal of Biology*, 2013)

Importance of Bioinformatics:

- Helps in retrieving DNA sequences.
- Helps in Sequence comparison.
- Helps in Gene function prediction.
- Helps in understanding Evolutionary relationship.
- Helps in creating Phylogenetic trees.

The tools that I used in this project were:

- **NCBI** → sequence retrieval
- **BLASTN** → sequence comparison
- **MUSCLE** → multiple sequence alignment
- **MEGA 11** → phylogenetic tree construction

3. Objectives

- Retrieve BRCA1 nucleotide sequences from NCBI.
- Compare BRCA1 sequences among mammalian species.
- Determine sequence similarity using BLASTN.
- Perform multiple sequence alignment using MUSCLE.
- Construct a phylogenetic tree using MEGA.
- Interpret the evolutionary relationship among species.

4. Materials and Methods

Database

- NCBI Gene
- NCBI RefSeq

Species

- Homo sapiens (National Center for Biotechnology Information, n.d.-c)
- Macaca mulatta (National Center for Biotechnology Information, n.d.-d)
- Canis lupus familiaris (National Center for Biotechnology Information, n.d.-b)
- Mus musculus (National Center for Biotechnology Information, 2026)
- Rattus norvegicus (National Center for Biotechnology Information, n.d.-a)

Software

- BLASTN (Camacho et al., 2009)
- MEGA 11
- MUSCLE (Edgar, 2004)

Method:

1. Sequence Retrieval and FASTA Download

In this study mRNA sequences of the BRCA1 gene were retrieve from NCBI gene and NCBI RefSeq database. For this five mammalian species (*Homo sapiens*, *Macaca mulatta*, *Canis lupus familiaris*, *Mus musculus* aur *Rattus norvegicus*) were selected, from which RefSeq accession number of BRCA1 mRNA sequence of each mammalian species was verified and the FASTA format of that sequence was downloaded. FASTA format was used as it is considered standard for DNA sequence analysis and easily read by bioinformatics software.

2. Pairwise BLAST Analysis

The downloaded sequences of mammalian species were compared by using BLASTN tool. Human BRCA1 sequence was used as query sequence and BRCA1 sequence of other mammalian species were used as subject sequence for comparison. After every comparison Percent Identity, Query Coverage and E-value was noted. The main purpose of BLASTN analysis was to evaluate the similarities in BRCA1 gene in different mammalian species.(Camacho et al., 2009)

3. Multiple Sequence Alignment

After completing BLAST analysis, all five BRCA1 mRNA sequences were combined into a single FASTA file. This file was imported to Mega 11 software and by using Muscle algorithm multiple sequence alignment was performed. Alignment helped us in identifying the conserved nucleotide regions and variable nucleotide position. This helped us in understanding the sequence similarities between different species and evolutionary differences.(Edgar, 2004)

4. Phylogenetic Tree Construction

After completing the Multiple sequence alignment, the aligned sequences were used in Neighnor-Joining(NJ) method in MEGA 11 software to construct phylogenetic tree. of Phylogenetic tree of selected mammalian specie helped me in visualizing and comparing the evolutionary relationship more clearly. (Felsenstein, 1985)

5. Bootstrap Analysis

We performed Bootstrap analysis for evaluating the reliability of phylogenetic tree. In this, 500 bootstrap replicates were used. The bootstrap vales are used to measure the statistical reliability of branches of phylogenetic tree. High bootstrap values indicates that the evolutionary relationships shown in phylogenetic tree is reliable and statistically well supported. (Felsenstein, 1985)

5. Results

- **Species and accession numbers.**

Species	Common Name	Accession Number
<i>Homo sapiens</i>	Human	NM_007294.4
<i>Macaca mulatta</i>	Rhesus macaque	NM_001114949.1
<i>Canis lupus familiaris</i>	Dog	NM_001013416.1
<i>Mus musculus</i>	Mouse	NM_009764.3
<i>Rattus norvegicus</i>	Rat	NM_012514.2

Table 1. Species and accession numbers of BRCA1 mRNA sequences used in this study.

- **BLAST comparison results.**

Comparison	Identity (%)	Query Coverage (%)	E-value
Human vs Monkey	96.03	79	0.0
Human vs Dog	83.97	79	0.0
Human vs Mouse	75.50	51	0.0
Human vs Rat	73.35	36	0.0

Table 2. Human BRCA1 showed the highest sequence similarity with *Macaca mulatta* (96.03%), followed by dog (83.97%), mouse (75.50%), and rat (73.35%). All comparisons produced an E-value of 0.0, indicating statistically significant sequence similarity.

- **Multiple sequence alignment.**

Representative multiple sequence alignment of BRCA1 mRNA sequences from *Homo sapiens*, *Macaca mulatta*, *Canis lupus familiaris*, *Mus musculus*, and *Rattus norvegicus* generated using the MUSCLE algorithm in MEGA 11. Conserved nucleotide positions are indicated by identical aligned bases, whereas nucleotide substitutions and alignment gaps reflect sequence variation and evolutionary divergence among the selected mammalian species.



Figure 2. Multiple sequence alignment of BRCA1 mRNA sequences from selected mammalian species.

- **Phylogenetic tree.**

The phylogenetic tree was constructed using the Neighbor-Joining method in MEGA 11 based on the multiple sequence alignment of BRCA1 mRNA sequences obtained from *Homo sapiens*, *Macaca mulatta*, *Canis lupus familiaris*, *Mus musculus*, and *Rattus norvegicus*. Bootstrap analysis was performed with 500 replicates to assess the reliability of the inferred relationships. Human and rhesus macaque clustered together, indicating the highest sequence similarity, while mouse and rat formed a distinct rodent clade. Dog occupied an intermediate evolutionary position between primates and rodents.



Figure 3. Neighbor-Joining phylogenetic tree showing the evolutionary relationship of BRCA1 mRNA sequences among selected mammalian species.

6. Discussion

1. Human and monkey clustered together due to close evolutionary relationship.

Homo sapiens and *Macaca mulatta* were clustered on a same branch on phylogenetic tree because of their close evolutionary relationship as they both are primates and evolved from the common ancestor. Due to this BRCA1 gene sequence showed most similarity (96.03% identity).

2. Mouse and rat formed a separate rodent clade.

Mus musculus, and *Rattus norvegicus* both are rodents due to which they made differ clade from primates in the phylogenetic tree. So, they have similarities in BRCA1 sequence according to evolutionary point of view represent same lineage.

3. Dog occupied an intermediate position.

Canis lupus familiaris was neither clustered with primates nor with rodents as it was in between both the groups on phylogenetic tree. This shows that BRCA1 sequence of dog is not as similar as monkey to humans but also it does not show that much evolutionary distance with mouse and rat. This supports the expected mammalian evolutionary pattern.

4. High bootstrap values indicate confidence in the inferred relationships.

Reliability of tree is shown by bootstrap value. In our case the bootstrap value of our tree was 100 which shows that software has supported the same grouping even after multiple analysis.

5. BRCA1 remains conserved across mammals.

BRCA1 is very important gene that is responsible for DNA repair, cell cycle regulation and maintains genome stability. Any mutation in this gene causes no DNA damage repair which lead to accumulation and uncontrol proliferation of cells, eventually causing the development of tumour. Due to this reason during evolution BRCA 1 was conserved by natural selection lead to the similarities in gene sequence in different mammals.(Burk-Herrick et al., 2006)

7. Conclusion

The results of this study demonstrate that the BRCA1 gene is highly conserved throughout mammalian evolution. Only a limited number of sequence variations were observed among the selected species, indicating that BRCA1 has remained largely unchanged over time due to its essential biological functions. Among the species analyzed, *Homo sapiens* and *Macaca mulatta* exhibited the highest sequence similarity. This finding reflects their close evolutionary relationship and supports the well-established classification of both species within the primate lineage. The use of bioinformatics tools, including NCBI, BLASTN, MUSCLE, and MEGA 11, enabled efficient retrieval, comparison, alignment, and phylogenetic analysis of BRCA1 sequences. These tools effectively revealed the evolutionary relationships among the selected mammalian species and demonstrated the value of computational approaches in comparative genomics and molecular evolution.

8. Future Scope

- Protein sequence comparison (BLASTP)
- Conserved domain analysis
- 3D protein structure prediction
- Motif analysis
- Functional mutation studies

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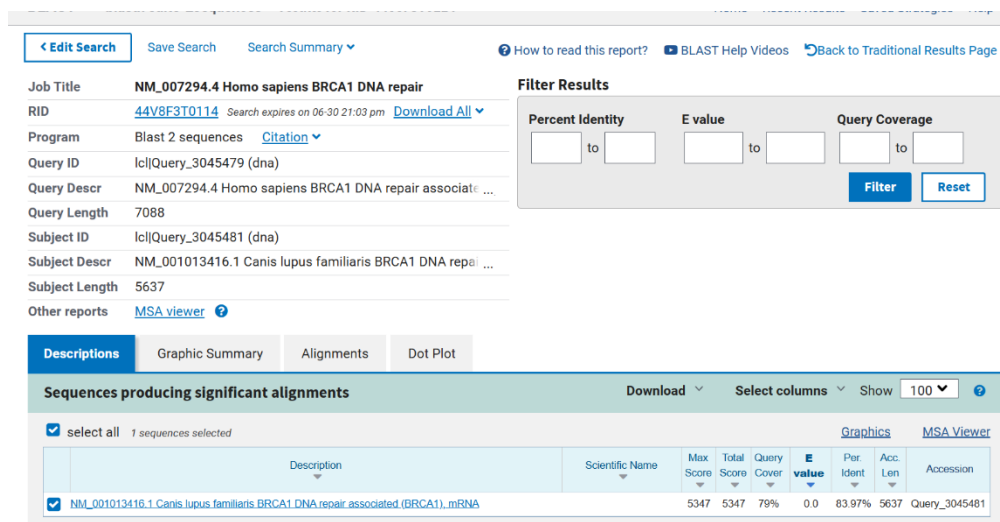
Zhong, A. X., Chen, Y., & Chen, P.-L. (2023). BRCA1 the Versatile Defender: Molecular to Environmental Perspectives. *International Journal of Molecular Sciences*, 24(18). <https://doi.org/10.3390/ijms241814276>

10. Appendix

- **Appendix A:** FASTA sequences:



- **Appendix B:** BLASTN Output Screenshots



- Figure B1: BLASTN comparison between *Homo sapiens* and *Canis lupus familiaris* BRCA1 mRNA sequences

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Query Descr NM_007294.4 Homo sapiens BRCA1 DNA repair associat...

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Subject Descr NM_009764.3 Mus musculus breast cancer 1, early onset ...

Subject Length 6648

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Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ NM_009764.3 Mus musculus breast cancer 1, early onset (Brca1), mRNA		1465	1762	51%	0.0	75.50%	6648	Query_275507

- Figure B2: BLASTN comparison between *Homo sapiens* and *Mus musculus* BRCA1 mRNA sequences.

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Query Descr NM_007294.4 Homo sapiens BRCA1 DNA repair associat...

Query Length 7088

Subject ID lclQuery_132203 (dna)

Subject Descr NM_012514.2 Rattus norvegicus BRCA1, DNA repair asso...

Subject Length 6651

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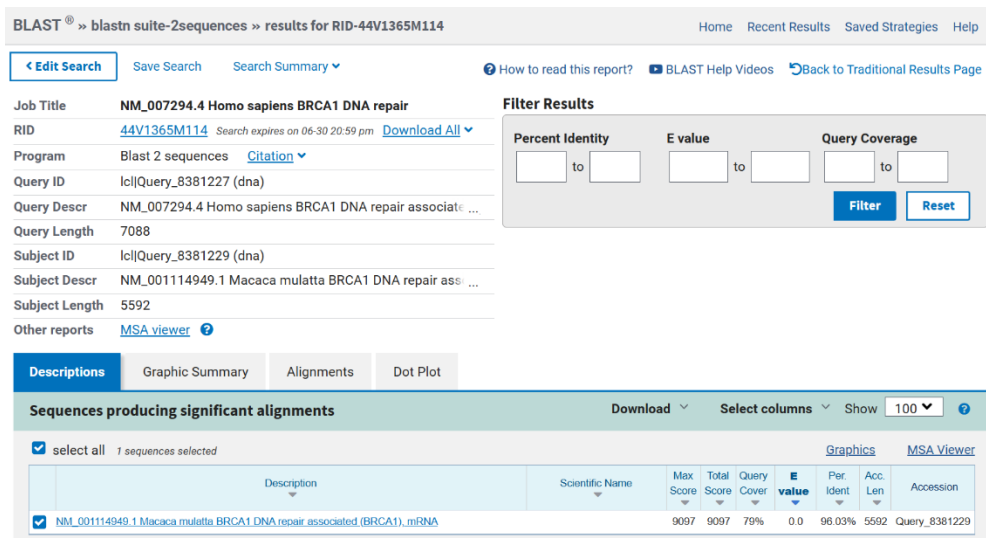
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Sequences producing significant alignments [Download](#) [Select columns](#) [Show](#) 100 [?](#)

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Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ NM_012514.2 Rattus norvegicus BRCA1, DNA repair associated (Brca1), mRNA		821	821	36%	0.0	73.35%	6651	Query_132203

- Figure B3: BLASTN comparison between *Homo sapiens* and *Rattus norvegicus* BRCA1 mRNA sequences.



- Figure B4: BLASTN comparison between *Homo sapiens* and *Macaca mulatta* BRCA1 mRNA sequences.

Appendix C: Alignment screenshots



Figure C1: Multiple sequence alignment of BRCA1 mRNA sequences from selected mammalian species.

Appendix D: Phylogenetic tree:



Figure D1: Neighbor-Joining phylogenetic tree showing the evolutionary relationship of BRCA1 mRNA sequences among selected mammalian species.